Sergio Estan-Ruiz

Email | Personal Website | GitHub | LinkedIn

Research Profile

PhD student at Imperial working on the mathematical foundations of AI and AI safety, with a particular focus on Singular Learning Theory and mechanistic interpretability.

Education

Imperial College London *2024–present*
PhD in Statistics and Machine Learning for Public Policy
- Supervised by Professor Guy Nason with funding from the competitive President's Scholarship.

University of Cambridge *2022–2023*
MASt in Mathematics- Part III
- Distinction (81%). Selected courses: Lie Algebras, Algebraic Topology, Differential Geometry, Advanced Probability, Stochastic Calculus, Statistical Learning Theory and Statistical Methods.
- Thesis (85%): *A Study of the Persistent Homology Pipeline in Data Analysis.*

University of Warwick *2019–2022*
BSc in Mathematics
- First Class Honours (92%, second highest in the cohort).
- Selected courses: Algebraic Topology, Manifolds, Probability Theory, Rings and Modules, Applied Stochastic Processes, Measure Theory, Commutative Algebra, and Partial Differential Equations.

Universidad Nacional de Educacion a Distancia (UNED) *2022–2026*
BSc in Economics, part-time
- Selected courses: Macroeconomics, Microeconomics, Econometrics and Public Budgeting.
- Thesis: *To What Extent is Japan Applying Modern Monetary Theory Doctrines?*

Research Experience

PhD Researcher *2024–present*
Imperial College London London, UK
- Designed forecasting, interpretable models for macroeconomic applications based on inferred network structures. Developed novel Bayesian and econometric tools to extend current network time series models.
- Investigated Singular Learning Theory as a lens on deep-learning behaviour to understand phenomena such as learning dynamics, grokking and refusal signals in language models.
- Explored topological methods for mechanistic interpretability in LLMs.
- Developed large-scale experimental pipelines in Python and PyTorch for model training, evaluation and distributed computation on HPC clusters.

Research Assistant *2023–2024*
IESE Business School Barcelona, Spain
- Research assistant to Dr. Diego Aparicio on online pricing and the existence of a pink tax.
- Used logistic fixed-effects models and developed Stata workflows for empirical analysis.

ICML Gold Reviewer *2026*
- Awarded to the top 25% of reviewers at ICML 2026.

Selected Programs

ILIAD Intensive at London Initiative for Safe AI (LISA) *2026*
- Month-long intensive programme on AI alignment, deep learning theory, interpretability and agent foundations.

BlueDot's Technical AI Safety Course *2026*
- echnical AI safety programme covering alignment, interpretability, evaluations and AI control.

Publications and Preprints

"GenGNAR: Genetic Graph Search for GNAR Models with Applications to Inflation Forecasting". *Sergio Estan-Ruiz**, Guy Nason*. Submitted. Pre-print at https://hdl.handle.net/10044/1/130611 — *2026*

"A Basin-Selection Perspective on Grokking via Singular Learning Theory". Ben Cullen*, *Sergio Estan-Ruiz**, Riya Danait, Jiayi Li. Submitted. Pre-print at https://arxiv.org/abs/2603.01192. — *2026*

Talks and Conferences

Invited Talks

An Introduction to AI Safety and Interpretability — *March/May 2026*
LLM and RL Imperial Seminar — Imperial, UK

Deep Learning Models for Latent Network Structure in MTS — *Sept./Oct. 2025*
Network Stochastic Processes and Time Series (NeST) Workshops — Oxford/Bath, UK

Contributed Talks

Open Problems in Dynamic Network-Based Time Series Models — *May 2025*
Random Network Workshop — Sheffield, UK

Graph-Based Time Series Models for Macroeconomic Forecasting — *April 2025*
International Mathematics and Statistics Student Research Symposium — Online

Poster

Graph-Based Macroeconomic Models — *June 2025*
Sunbelt Conference, International Network for Social Network Analysis — Sorbonne, France

Teaching and Academic Service

Co-founder of LLM and Reinforcement Learning Seminar Series — *2026*
Imperial College London — London, UK

- Jointly founded and organised with Paul Lezeau the Imperial LLM and RL seminar, coordinating talks from researchers across Imperial, industry and external institutions.

Graduate Teaching Assistant — *2024–2026*
Imperial College London — London, UK

- Taught BSc and MSc courses, including Linear Algebra, Numerical Methods, Introduction to Statistical Finance, Network Science, Fundamentals of Statistical Inference and Deep Learning.

Joint MSc Thesis Supervisor — *2025*
Imperial College London — London, UK

- Jointly supervised, with Professor Guy Nason and Dr Lloyd Fung, an MSc thesis on discovering equations.

Mathematics Tripos Supervisor and STEP Examiner — *2023–2024*
University of Cambridge — Cambridge, UK

- Supervised second-year modules for students across nearly 10 colleges, including Variational Principles, Linear Algebra, and Numerical Analysis.
- Contributed to mark-scheme development and marking for STEP, the entrance exam for the Cambridge Mathematics Tripos.

Awards and Funding

ICML Gold Reviewer — *2026*

- Awarded to the top 25% of reviewers at ICML 2026.

President's PhD Scholarship — *2024–2028*

- Competitive award supporting doctoral study at Imperial College London.

International Mathematics Competition — *2022*

- Second award as a selected team member for the University of Warwick.

Short Term Mobility Bursary — *2022*

- Bursary supporting research in representation theory at the University of Toulouse.

COBIS Excellence Award and Pearson Outstanding Learner Award *2019*

- Awarded by the Council of British International Schools and Pearson for academic excellence and outstanding GCE results, including the highest marks in Mathematics and Chemistry in Europe and Spain respectively.

Sir Roger Fry Scholarship *2017–2019*

- Scholarship of 10,000 EUR to pursue sixth form at King's College Alicante.

Technical Skills

Programming: Python (PyTorch, Ripser), R (GNAR, igraph, parallel), Stata
Scientific Computing: HPC clusters, GPU computing, parallel experiments
Tools: Git, LaTeX and basic Linux
Languages: English, Spanish, Catalan (Proficient); French (Advanced)